Climate Shocks, Health Finance, and Economic Outcomes

County-Level Evidence from the United States, 2011–2023

Thesis Committee Update · March 2026

Project Background

Climate, health finance, and economic outcomes in the U.S., 2011–2023

Research Question

- Do climate shocks (cold, heat, drought, air quality) affect the **cost and accessibility** of health care?
- Do they propagate into **household economic outcomes** — income, debt, employment?

Motivation

- Climate adaptation policy needs **health-finance channel** evidence
- Literature centres on mortality; **cost & debt channels understudied**
- U.S. county two-way FE enables within-unit identification

Data Coverage

Domain	Source
Climate (Temp, HDD, CDD)	NOAA/NCEI
Drought index (PDSI)	NOAA/NCEI
Air quality (AQI)	EPA
HIX premiums	HIX Compare
Medical debt	Urban Institute
Hospital bad debt	NASHP HCT
Employer insurance	AHRQ MEPS-IC
Public payers	CMS NHE
HH income / employment	ACS / BEA

Panel: 2011–2023; ∼3,155 counties; state panel: 51 states × 13 years.

Empirical Specification

Two-way fixed effects with distributed lags; all dollar outcomes inflation-adjusted to $2023

Estimating Equation

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^{2} \beta_h \, \text{Shock}_{i,t-h} + \mathbf{X}'_{it}\delta + \varepsilon_{it}$$

- α_i: unit FE (state or county); γ_t: year FE
- Distributed lags $h = 0, 1, 2$ for all shocks
- SE clustered at **state level** throughout
- **County:** also rating-area-clustered SEs for premiums

Climate Shock Variables

Variable	Definition
`High_CDD`	CDD $\geq$ national p80 (1990–2000)
`High_HDD`	HDD $\geq$ national p80 (1990–2000)
`Z_Temp`	$(T_{it} - \bar{T}_i^{\text{90-00}})/\sigma_i^{\text{90-00}}$
`Z_Precip`	Precipitation z-score (same baseline)
PDSI	Palmer drought severity index
`High_AQI_Max`	Max AQI $>$ 100 (binary)

Outcome Blocks

Outcome	Source
Employer premium contribution	MEPS-IC
Benchmark Silver premium	HIX
Medical debt share / median	Urban Inst.
Hospital bad debt per capita	NASHP
Total, Medicaid, Medicare spending	CMS
Median HH income / employment	ACS
Per-capita personal income	BEA

Dynamic Estimation Strategy

Baseline dynamic-panel distributed lags and local projections recover complementary timing margins

Dynamic Panel / Distributed Lag (DP/DL)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^{2} \beta_h \,\mathrm{Shock}_{i,t-h} + \mathbf{X}_{it}'\delta + \varepsilon_{it}$$

- Main FE estimator for contemporaneous and lagged effects
- Cumulative response is the sum of lag coefficients
- Baseline estimator for the state and county tables

Local Projections (LP, Jordà 2005)

$$Y_{i,t+h} = \alpha_i^h + \gamma_t^h + \beta_h \,\mathrm{Shock}_{i,t} + \mathbf{X}_{it}'\delta_h + \varepsilon_{i,t+h}$$

- Separate regression for each horizon $h \in \{-2, \ldots, +3\}$
- Negative horizons serve as placebo leads for pre-trend checks
- More flexible when lag shapes are misspecified

Controls in both approaches: uninsured rate, median household income, and the same county/year fixed effects.
Inference: state-clustered SEs throughout; rating-area clustered variants for premium outcomes as robustness.

State-Level Results

Two-way FE: State + Year, $N \approx 550$–650, cluster = State

State: Drought and Cold Raise Premiums and Debt

Household & individual outcomes; $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$

Employer Premium Contribution ($/yr)

Predictor	Coef.	
Extreme Drought (h=0)	+63.6	**
Severe Drought (h=0)	+86.4	***
Severe Drought (lag 1)	+65.1	**
Pct $PM_{2.5}$ days	+20.8	***
Pct PM_{10} days	+22.7	***
Pct Ozone days	+19.7	***

Air quality pollutant days show consistent positive effects on employer premium costs.

Medical Debt Share (pp)

Predictor	Coef.	
Cold Shock (lag 1)	+0.0135	**

Medical Debt Median ($/yr)

Predictor	Coef.	
High HDD (h=0)	+58.9	**
Severe Drought (h=0)	−66.8	***
Severe Drought (lag 1)	−37.5	**
Heat Shock (h=0)	+32.7	*
Heat Shock (lag 2)	+47.7	**

Cold raises debt prevalence; drought paradoxically reduces median severity.

State: Cold Raises System Costs; Drought Cuts Public Payer Spending

Hospital & system outcomes; two-way FE (State + Year)

Total Per-Capita Health Expenditure ($)

Predictor	Coef.	
Cold Shock (h=0)	+205.3	**
High HDD (lag 2)	+92.0	*

Medicaid per Enrollee ($)

Predictor	Coef.	
Extreme Drought (h=0)	−566.2	**
Severe Drought (h=0)	−525.8	**
Severe Drought (lag 1)	−524.5	***
Severe Drought (lag 2)	−288.4	**

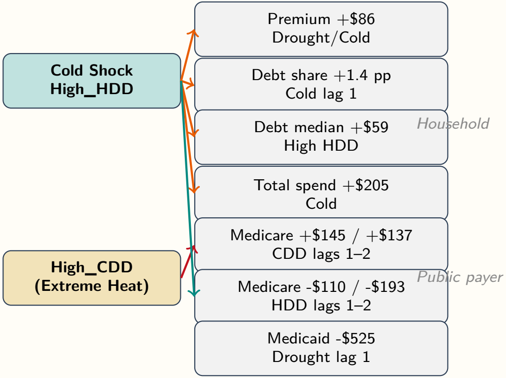
Drought persistently reduces Medicaid spending across lags 0–2.

Medicare per Enrollee ($)

Predictor	Coef.	
Severe Drought (lag 1)	−144.3	***
High CDD (lag 1)	+144.7	**
High CDD (lag 2)	+137.2	***
High HDD (lag 1)	−110.1	*
High HDD (lag 2)	−192.7	***
Unhealthy days	+97.7	**

Heat (CDD) raises Medicare with lag; cold (HDD) reduces it — opposite channels.

State-Level: Cold and Heat Drive Opposite Financial Channels



Cold channel:

- Premiums, debt, and total spending rise
- Medicare falls with lagged HDD exposure

Heat channel (CDD):

- Medicare rises 1–2 years later

Drought:

- Medicaid falls and premiums rise

Heat and cold affect public payers in opposite directions — novel finding.

County Results: Household & Individual

Medical debt, premiums, income, employment
Two-way FE: County + Year, $N \approx 23k–34k$, cluster = State

County: Heat Raises Debt Severity; Cold Raises Debt Prevalence

Medical Debt Share (pp) and Median Debt ($2023); UW = unweighted, PW = pop-weighted

Z_Temp (Relative Heat Shock)

	UW	PW
Debt Share (pp)		
h=0	−0.0025*	−0.0000
Lag 2	−0.0027**	−0.0014*
Debt Median ($)		
h=0	+14.9*	+18.5***
Lag 2	+19.5**	+16.2***

Heat reduces prevalence but raises severity — consistent composition effect: warm years shed marginal debtors while pushing the remaining cases deeper.

High_HDD (Extreme Cold Burden)

	UW	PW
Debt Share (pp)		
h=0	+0.0038**	+0.0054**
Lag 2	+0.0031*	+0.0037*
Debt Median ($)		
h=0	+10.6	+31.2**

Controls (all debt models):

Uninsured Rate	+*** **(large)**
HH Income	−**

Cold raises both prevalence and severity — no composition effect.

County: Cold Raises Premiums Now; Heat Raises Them Later

Benchmark Silver Premium ($2023 real); county + year FE

High_HDD (Cold Burden)

Lag	UW	PW	Cluster
h=0	+28.1***	+21.9**	State
h=0	+28.1***	+21.9***	RA
Lag 1	−24.5*	−20.8	State

High_CDD (Heat Burden) — lagged

Lag	UW	PW	Cluster
Lag 1	+8.7	+31.4***	RA
Lag 2	+21.6*	+35.3***	RA

Z_Temp (Relative Temp)

Lag	UW	PW	Cluster
h=0	−10.9**	+2.7	RA
Lag 1	+3.8	+5.4**	RA
Lag 2	+1.0	+4.2**	RA

- **Cold (HDD)** raises premiums on impact in both state- and RA-clustered specs
- **Heat** lowers premiums now, then raises them 1–2 years later
- RA clustering tightens CIs; drought lag 1 is borderline at +$3.9 ($p$=0.05)

County: Heat Depresses Income; Drought Hits PCPI

Median HH income and per-capita personal income ($2023)

Median HH Income ($2023)

Predictor	UW	PW
Z_Temp (h=0)	−71**	−88***
Z_Temp (lag 1)	−36	−64***
Z_Temp (lag 2)	−71**	−60**
High_CDD (h=0)	−247**	−3
High_HDD (h=0)	+161**	+173***

Relative

heat most consistently lowers household income in the population-weighted specifications.

Per-Capita Personal Income ($)

Predictor	UW	PW
PDSI lag 2	−8.5	−113**
High_CDD lag 1	+480**	−116
RA-level: PDSI (h=0)	−206***	

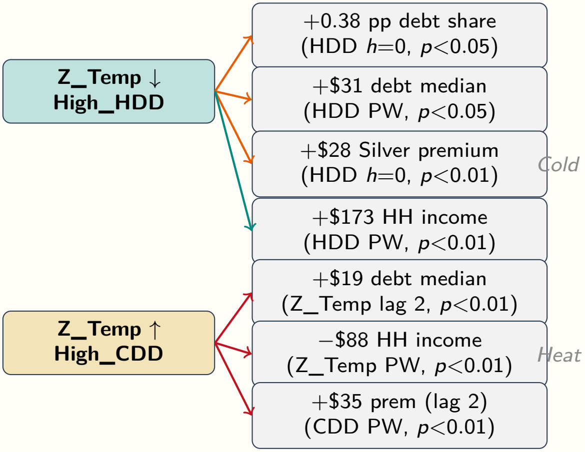
Income Summary

- **Heat** lowers HH income across lags in pop-weighted models
- **Cold** raises HH income on impact, likely reflecting seasonal or energy-income composition
- **Drought** is strongest in PCPI: RA spec gives −$206***

Civilian Employment (persons)

Predictor	UW	PW
Z_Temp lag 2	+788**	+3216
High_HDD (h=0)	−468**	+2587
High_HDD lag 2	−678**	+3988
Z_Precip (h=0)	+46	−5742***
Z_Precip (lag 1)	−88	−3803***

County-Level: Cold and Heat Channels Mirror the State Findings



Cold channel:

- Raises debt prevalence and severity
- Raises premiums on impact
- Raises HH income (energy mix?)

Heat channel:

- Raises debt median (lagged)
- **Reduces** HH income contemporaneously
- Raises premiums with 1–2 yr lag

Cross-level:

- Cold → debt: both levels confirmed
- Heat → income: county only
- Premium channels align in direction

County Results: Hospital Sector

Hospital bad debt per capita

Two-way FE: County + Year, $N \approx 27k$, cluster = State

County: Cold Raises Hospital Bad Debt; Heat Lag Reduces It

Hospital Bad Debt per Capita ($); within-$R^2 = 0.006–0.019$

Key Coefficient Estimates

Predictor	UW	PW
Z_Temp lag 2	-2.41^{***}	-2.08^{***}
Z_Temp (h=0)	-1.25	-1.08^{**}
High_HDD (h=0)	$+5.33^{***}$	$+4.23^{**}$
PDSI lag 1	$+0.57^{*}$	$+0.50^{*}$
PDSI lag 2	$+0.01$	$+0.50^{*}$
Uninsured Rate	$+151^{***}$	$+142$

RA-level robustness (pop-wtd):

PDSI lag 2: $+0.76^{***}$ Z_Temp lag 2: -2.26^{***}

- **Cold (HDD) raises bad debt on impact** by $4–5 per capita — consistent with emergency utilisation, deferred billing, and coverage gaps during cold events
- **Heat lag reduces bad debt** (2-yr lag, $p<0.001$) — possible reduction in elective utilisation during hot years or survivor selection
- **PDSI (drought)** adds a smaller lagged increase, confirmed in RA robustness
- **Uninsured Rate** is the dominant control ($+151^{***}$ unweighted) — structural coverage gap drives hospital write-offs

About 23% of county-years lack hospital reports; one negative outlier is winsorized.

Cross-Level Synthesis

What do state and county results agree on?

State and County Pipelines Tell a Consistent Story

Key findings replicated across both levels of analysis

Finding	State	County
Cold / HDD → medical debt (prevalence & severity)	✓	✓
Cold / HDD → higher premiums	✓ (via drought)	✓ ($p<0.01$)
Cold / HDD → hospital bad debt ↑	—	✓ ($p<0.01$)
Drought → reduced Medicaid / PCPI	✓ ($p<0.01$)	✓ (PDSI)
Heat → household income loss	—	✓ (Z_Temp***)
Heat lag → Medicare / premiums ↑	✓ (CDD***)	✓ (CDD***)
Heat lag → hospital bad debt ↓	—	✓ ($p<0.001$)
Poor air quality → higher premiums	✓ ($PM_{2.5}/O_3$***)	not modeled

Model Diagnostics and Robustness

Both pipelines pass standard specification checks

State-Level

- $N = 550$–650; Within-$R^2 = 0.09$–0.21
- `Pct_NO2` dropped by `fixest` (collinear)
- VIF max ≈ 5.3 after PDSI-only drought block
- CDD/HDD use national p80 thresholds from the 1990–2000 baseline
- `is_high_hdd` added in the current draft

County-Level

- $N = 23\text{k}$–34k; Within-R^2 up to 0.81
- RA-clustered SEs tighten CIs; core results survive
- RA robustness re-estimates the premium lags with pop weights
- CO 2023 debt observations excluded under HB23-1126 (60 obs)
- Z-scores: 1990–2000 per-county baseline
- Hospital data: $\sim$23% missing counties

Full tables: `state_regression_results.md` and `county_regression_results.md`

Next Steps

Planned extensions to the empirical framework

Empirical Extensions

- **IRFs:** extend LP profiles for `High_CDD`, `High_HDD`, drought, and AQI through $h = 4$
- **County unemployment:** replace the ACS proxy with BLS LAUS
- **Mechanisms:** separate respiratory-illness from energy-poverty channels for cold
- **AQI county models:** add AQI shocks to premium and debt specifications

Writing and Defence

- **Chapter draft:** unify the state and county narratives
- **Heterogeneity:** rural vs. urban and high- vs. low-AQI counties
- **Policy counterfactuals:** ACA Medicaid expansion $\times$ climate shocks

All pipeline code is version-controlled; datasets are documented in `CLAUDE.md`.